

Lab 11: Clustering Analysis with K-means, Hierarchical Clustering, and DBSCAN

Objective:

- Hands-on practice with three clustering algorithms—K-means, Agglomerative (Hierarchical) Clustering, and DBSCAN—applied to a real-world customer segmentation dataset. You will load, preprocess, cluster, evaluate, tune hyperparameters, and compare results side by side.

Software Requirements:

- (1) Python
- (2) scikit-learn
- (3) pandas
- (4) numpy
- (5) matplotlib
- (6) scipy

Dataset:

You will use the Wholesale Customers Data Set for this clustering analysis. This dataset contains the annual spending amounts of clients in diverse product categories, such as Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicassen. It also includes information such as Channel and Region. The dataset is suitable for customer segmentation and clustering analysis, where the aim is to identify groups of customers with similar purchasing patterns.

Lab Instructions

1. Import necessary libraries.

2. Follow the exercises below to perform clustering analysis using K-means, Hierarchical Clustering, and DBSCAN.

- 1) Data Loading and Exploration: (10pts)
 - Use Pandas to load the dataset.
 - Display the first 5 rows of the dataset.
 - Explore the dataset using methods like head(), describe(), info().
 - Check for missing values.
 - Visualize distributions of key features using histograms or boxplots.
- 2) Data Preprocessing: (15pts)
 - Check the dataset columns and confirm which columns will be used for clustering.
 - Handle categorical-like variables appropriately if needed (e.g., Channel and Region).
 - Extract features from the selected columns and standardize the features to zero mean and unit variance.

- 3) Model Building: (30pts)
 - 3.1) K-means Clustering
 - Fit K-means with k=5.
 - Record clustering labels and fit time.
 - 3.2) Hierarchical Clustering
 - Plot a dendrogram using Ward's linkage to inspect cluster structure.
 - Fit AgglomerativeClustering with n_clusters=5, linkage='ward'.
 - 3.3) DBSCAN Clustering
 - Fit DBSCAN with eps=0.5, min_samples=5.
 - Record clustering labels (note: -1 indicates noise) and fit time.
- 4) Model Evaluation: (15pts)
 - Define a function to compute Silhouette Score only when at least two non-noise clusters are present (for DBSCAN, exclude points labeled as -1).
 - Compute and tabulate: Silhouette Score, Fit time and Number of clusters (especially for DBSCAN).
- 5) Hyperparameter Tuning: (25pts)
 - Systematically explore different parameter combinations for each clustering algorithm and select the best setting based on Silhouette Score when valid. You should take into account the following parameters or settings.
 - Plot the clusters for each method under the optimal parameters (use PCA for dimensionality reduction if necessary).
 - 5.1) K-means Clustering: k
 - 5.2) Hierarchical Clustering: linkage and k
 - 5.3) DBSCAN Clustering: eps and min_samples
- 6) Performance Comparison: (5pts)
 - Summarize best configuration for each algorithm in a table, including best parameter and Silhouette.
 - Plot a bar chart of best Silhouette Scores for three methods.

Submission Requirement:

- 1) Complete all sections in a jupyter notebook file, name your file as "name_student number.ipynb" and submit it on iSpace.
- 2) Please submit the executed code.

Deadline: 23:59, May.31, 2026; Plagiarism (cheating) = F